

Supporting Materials for LONSA

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1 Supporting Materials for LONSA

$$\begin{aligned}
 A_L &= \sum_{l=1}^K \sum_{k=l}^K (n_k - 1) (n_{k+1} + n_{k+2} + \cdots + n_K) \\
 &= \underbrace{\sum_{l=1}^K \sum_{k=l}^K n_k (n_{k+1} + n_{k+2} + \cdots + n_K)}_{X_L} - \underbrace{\sum_{l=1}^K \sum_{k=l}^K (n_{k+1} + n_{k+2} + \cdots + n_K)}_{Y_L} \\
 &= \sum_{k=1}^{K-1} k n_k (n_{k+1} + n_{k+2} + \cdots + n_K) - \frac{1}{2} \sum_{k=2}^K k(k-1) n_k \quad \text{From Equation (2), (3)} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 X_L &= \sum_{l=1}^K \sum_{k=l}^K n_k (n_{k+1} + n_{k+2} + \cdots + n_K) \\
 &= [n_1 (n_2 + n_3 + \cdots + n_K) + n_2 (n_3 + n_4 + \cdots + n_K) + \cdots + n_{K-1} (n_K)] + \\
 &\quad [n_2 (n_3 + n_4 + \cdots + n_K) + n_3 (n_4 + n_5 + \cdots + n_K) + \cdots + n_{K-1} (n_K)] + \\
 &\quad [n_3 (n_4 + n_5 + \cdots + n_K) + n_4 (n_5 + n_6 + \cdots + n_K) + \cdots + n_{K-1} (n_K)] + \\
 &\quad \cdots + [n_{K-1} (n_K)] \\
 &= n_1 (n_2 + n_3 + \cdots + n_K) + 2n_2 (n_3 + n_4 + \cdots + n_K) + \cdots + (K-1)n_{K-1} (n_K) \\
 &= \sum_{k=1}^{K-1} k n_k (n_{k+1} + n_{k+2} + \cdots + n_K) \quad (2)
 \end{aligned}$$

$$\begin{aligned}
Y_L &= \sum_{l=1}^K \sum_{k=l}^K (n_{k+1} + n_{k+2} + \cdots + n_K) \\
&= [(n_2 + n_3 + \cdots + n_K) + (n_3 + n_4 + \cdots + n_K) + \cdots + (n_K)] + \\
&\quad [(n_3 + n_4 + \cdots + n_K) + (n_4 + n_5 + \cdots + n_K) + \cdots + (n_K)] + \\
&\quad [(n_4 + n_5 + \cdots + n_K) + (n_5 + n_6 + \cdots + n_K) + \cdots + (n_K)] + \cdots + \\
&\quad [(n_K)] \\
&= [n_2 + 2n_3 + \cdots + (K-1)n_K] + [n_3 + 2n_4 + \cdots + (K-2)n_K] + \\
&\quad [n_4 + 2n_5 + \cdots + (K-3)n_K] + \cdots + [(n_K)] \\
&= n_2 + (2+1)n_3 + (3+2+1)n_4 + (4+3+2+1)n_5 + \cdots + \\
&\quad ((K-1) + (K-2) + \cdots + 1)n_K \\
&= \sum_{k=2}^K [(k-1) + (k-2) + \cdots + 1] n_k \\
&= \frac{1}{2} \sum_{k=2}^K k(k-1)n_k
\end{aligned} \tag{3}$$

$$\begin{aligned}
B_L &= \frac{1}{2} \sum_{l=1}^K \sum_{k=l}^K n_k(n_k - 1) \\
&= \frac{1}{2} \left[\sum_{l=1}^K \sum_{k=l}^K (n_k^2 - n_k) \right] \\
&= \frac{1}{2} \left[\sum_{l=1}^K \sum_{k=l}^K n_k^2 \right] - \frac{1}{2} \left[\sum_{l=1}^K \sum_{k=l}^K n_k \right] \\
&= \frac{1}{2} [(n_1^2 + n_2^2 + \cdots + n_K^2) + (n_2^2 + n_3^2 + \cdots + n_K^2) + \cdots + (n_K^2)] - \\
&\quad \frac{1}{2} [(n_1 + n_2 + \cdots + n_K) + (n_2 + n_3 + \cdots + n_K) + \cdots + (n_K)] \\
&= \frac{1}{2} (n_1^2 + 2n_2^2 + 3n_3^2 + \cdots + Kn_K^2) - \frac{1}{2} (n_1 + 2n_2 + 3n_3 + \cdots + Kn_K) \\
&= \left[\frac{1}{2} \sum_{k=1}^K kn_k^2 \right] - \left[\frac{1}{2} \sum_{k=1}^K kn_k \right] \\
&= \frac{1}{2} \sum_{k=1}^K kn_k(n_k - 1)
\end{aligned} \tag{4}$$

$$\begin{aligned}
C_L &= \sum_{l=1}^K \sum_{k=l+1}^K n_k \\
&= (n_2 + n_3 + \cdots + n_K) + (n_3 + n_4 + \cdots + n_K) + \cdots + (n_K) \\
&= n_2 + 2n_3 + 3n_4 + \cdots + (K-1)n_K \\
&= \sum_{k=2}^K (k-1)n_k
\end{aligned} \tag{5}$$

$$\begin{aligned}
dC &= \sum_{k=1}^{K-1} kZ(K-k)Z - \frac{1}{2} \sum_{k=2}^K (k-1)(k-2)Z + \frac{1}{2} \sum_{k=1}^K kZ(Z-1) \\
&= Z^2 \sum_{k=1}^{K-1} k(K-k) - \frac{1}{2} Z \sum_{k=1}^K (k^2 - 3k + 2) + \frac{1}{2} Z(Z-1) \sum_{k=1}^K k \\
&= Z^2 K \sum_{k=1}^{K-1} k - Z^2 \sum_{k=1}^{K-1} k^2 - \frac{1}{2} Z \sum_{k=2}^K k^2 + \frac{3}{2} Z \sum_{k=1}^K k - Z \sum_{k=1}^K 1 + \frac{1}{4} N(Z-1)(K+1) \\
&= NZ \frac{1}{2} K(K-1) - Z^2 \frac{1}{6} (K-1)K(2K-1) - \frac{1}{2} Z \frac{1}{6} K(K+1)(2K+1) + \\
&\quad \frac{3}{2} Z \frac{1}{2} K(K+1) - ZK + \frac{1}{4} N(Z-1)(K+1) \\
&= \frac{1}{2} N^2(K-1) - \frac{1}{6} NZ(K-1)(2K-1) - \frac{1}{12} N(K+1)(2K+1) + \\
&\quad \frac{3}{4} N(K+1) - N + \frac{1}{4} N(KZ + Z - K - 1) \\
&= \frac{1}{2} N^2(K-1) - \frac{1}{6} NZ(2K^2 - 3K + 1) - \frac{1}{12} N(2K^2 + 3K + 1) + \\
&\quad \frac{3}{4} N(K+1) - N + \frac{1}{4} N(N + Z - K - 1) \\
&= \frac{1}{12} [6N^2K - 6N^2 - 4NZK^2 + 6NZK - 2NZ - 2NK^2 - 3KN - N + \\
&\quad 9NK + 9N - 12N + 3N^2 + 3NZ - 3NK - 3N] \\
&= \frac{1}{12} [6N^2K - 6N^2 - 4N^2K + 6N^2 - 2NZ - 2NK^2 - 3KN - N + \\
&\quad 9NK + 9N - 12N + 3N^2 + 3NZ - 3NK - 3N] \\
&= \frac{1}{12} [(6N^2K - 4N^2K) + (-2NZ + 3NZ) - 2NK^2 + \\
&\quad (-3KN - 3NK + 9NK) + (-N + 9N - 12N - 3N) + 3N^2] \\
&= \frac{1}{12} (2N^2K + NZ - 2NK^2 + 3NK - 7N + 3N^2) \\
&= \frac{1}{12} N (2NK - 2K^2 + 3K - 7 + 3N + Z)
\end{aligned} \tag{6}$$